

Phylogenetic Analysis on the *rbcL* Gene Using the DNA of Four Distantly Related Algal Species

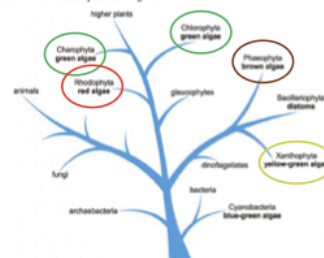
Xochitl Ramirez

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Dataset Selection

Although all are algae, they come from different phyla, have different morphologies, evolved at different points in time, yet all contain the *rbcL* gene

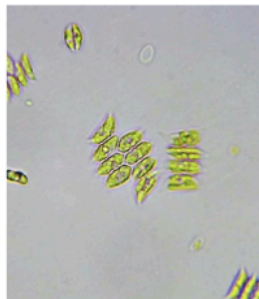


Vaucheria litoria
Coenocytic, Yellow-green

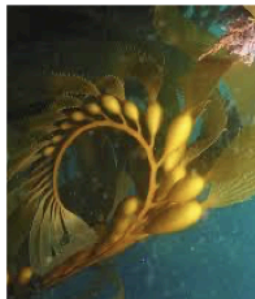


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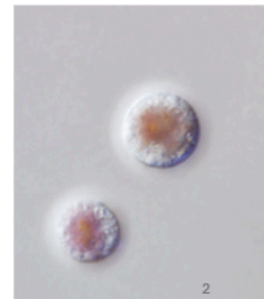
Tetrademus obliquus
Colonial, Green



Macrocystis pyrifera
Multicellular, Brown



Porphyridium purpureum
Unicellular, Red



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Visualization of Nucleic Acids

Tool: N•ESPrpt 0.9 - Webversion, NESPrpt.cgi 4.1.5

Reason: Evaluates biochem properties found in aligned sequences to check if chem properties of bases changed or stayed the same .

Observation: Beg & end of sequences differ but mid sections match well. Reveals this part of the gene is conserved amongst all four species.

KEY:

- Red Blocks, White Letters: Identical
- Blue-Outlined Box, Red Letters: Similarity above 70%, same chemical group
- No Box, Black Letters: Low similarity, below 70%, different chemical group

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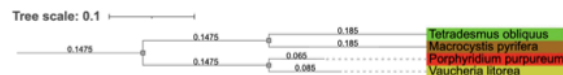
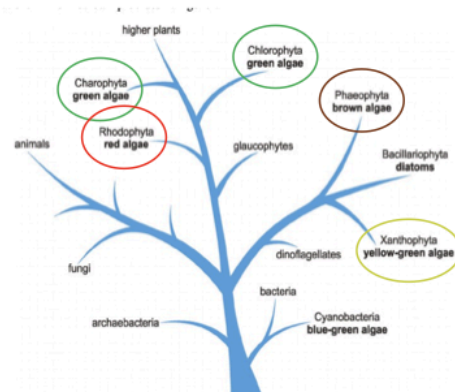
Phylogenetic Analysis

Tool: Interactive Tree of Life (iTOL) - Version 7.6

Reason: Free, publicly available web program with easy-to-use interface and effectively maps evolutionary distance between organisms

Observation:

- Result: *V. litorea* (YG) is closest *P. purpureum* (R), while *T. obliquus* (G) is closest to *M. pyrifera* (B) – contradiction of traditional TOL.
- Reason: Strict focus on the *rbcL* gene. YG algae evolved from primitive organism that engulfed and absorbed organelles of Red algae into its cellular structure, causing chloroplast to become part of the YG algae. Thus, chloroplast closely related.



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Synthesis

What can be learned?

Studying algal DNA can help us better understand gene conservation and organismal divergence by unmasking the effects of genetic redundancy that protein sequencing creates.

Where can uncertainty be introduced?

Relying on a single gene (*rbcL*) can introduce single-gene bias (oversimplification of complex traits), which can create a tree that contradicts established organismal phylogeny.

How can uncertainty be mitigated?

To mitigate single-gene bias, phylogenetic work should focus on analyzing multiple conserved genes from several organisms.

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Report (w/
References),
Notes, & Appendix
Stuff

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